

Driver Quality & Service Performance Analytics

1. Executive Summary

This project analyzes ride-booking and service-performance data to identify patterns associated with ride completion, cancellations, vehicle arrival time, pickup location, and customer experience.

The dataset contains **150,000 bookings**, of which **93,000 were completed**, resulting in an overall completion rate of **62.0%**. The remaining **38.0%** represent non-completed bookings, including driver cancellations, customer cancellations, incomplete rides, and cases where no driver was found.

The analysis shows that **pickup location and vehicle arrival time are the most notable dimensions of performance variation**. Vinobapuri recorded a non-completion rate of 45.32%, compared with 33.73% in Welcome. Customer cancellation rates also increase substantially across VTAT bands.

In comparison, vehicle type and broad time periods show relatively small differences in completion performance.

The project combines **Python, PostgreSQL, SQL, and Power BI** to transform raw booking data into operational insights and an interactive reporting dashboard.

2. Project Objective

The objective is to understand ride-service performance from an operational perspective and answer the following questions:

- What proportion of bookings are successfully completed?
- Where are non-completion and cancellation rates highest?
- What are the main driver-cancellation reasons?
- Is vehicle arrival time associated with customer cancellations?
- Do vehicle types show meaningful performance differences?
- Are there important time-based patterns?
- What do driver and customer ratings indicate about service quality?

The analysis is descriptive and identifies patterns and associations rather than causal relationships.

3. Data Source

The dataset was obtained from Kaggle:

<https://www.kaggle.com/datasets/anilrohan/uber-data-india?resource=download>

The raw dataset and derived CSV files are maintained locally. The analysis notebook recreates the cleaned and feature-engineered dataset from the original source.

The cleaned dataset was subsequently loaded into PostgreSQL for SQL-based analysis and Power BI reporting.

4. Data Preparation

4.1 Data Cleaning

The dataset was reviewed for:

- Missing values
- Duplicate rows and Booking IDs
- Invalid categorical values
- Rating ranges
- Numerical field consistency
- Booking-status consistency

Missing values were retained when they were structurally not applicable. For example, ratings may not exist for cancelled bookings because no completed ride occurred.

4.2 Feature Engineering

Additional variables were created to support operational analysis:

Temporal features

- Month
- Day of Week
- Hour
- Time Period

- Is Weekend

Booking outcome indicators

- Is Completed
- Is Driver Cancelled
- Is Customer Cancelled
- Is Incomplete

Segmentation features

- Driver Rating Band
- VTAT Band
- CTAT Band
- Distance Band

These features were used throughout the SQL analysis and Power BI dashboard.

5. Dataset Overview

Metric	Result
Total bookings	150,000
Completed rides	93,000
Completion rate	62.0%
Non-completion rate	38.0%
Driver cancellations	27,000
Customer cancellations	10,500
Incomplete rides	9,000
No driver found	10,500
Pickup locations	176

The **62.0% completion rate** provides the baseline for comparing different locations, vehicle types, time periods, and service conditions.

6. Key Business Findings

6.1 Overall Booking Performance

Out of 150,000 bookings, **93,000 were completed**, giving a completion rate of **62.0%**.

Driver cancellations are the largest individual non-completion category at **18.0% of all bookings**.

This indicates that cancellation and non-completion behavior represents a significant part of overall booking performance and should be monitored alongside completed rides.

6.2 Pickup Location Performance

Pickup location shows noticeable variation in non-completion rates.

Pickup Location	Non-Completion Rate
Vinobapuri	45.32%
Akshardham	43.86%
Welcome	33.73%

The difference between Vinobapuri and Welcome is approximately **11.6 percentage points**.

Insight

Location is therefore an important operational segmentation variable. High non-completion locations can be investigated further for differences in driver availability, cancellation behavior, pickup conditions, and customer demand.

The available dataset does not contain enough contextual information to determine the underlying cause.

6.3 Driver Cancellation Reasons

Driver-cancellation reasons are relatively evenly distributed.

Reason	Share of Driver Cancellations
Customer-related issue	25.32%
Coughing / sickness	25.00%
Personal / car-related issue	24.91%
More passengers than permitted	24.76%

Insight

No single cancellation reason dominates the dataset. Therefore, cancellation reasons should be analyzed together with location, time, and other operational dimensions rather than in isolation.

6.4 VTAT and Customer Cancellations

Customer cancellation rates vary substantially across VTAT bands.

VTAT Band	Customer Cancellation Rate
Fast	0.11%
Moderate	5.86%
Slow	7.68%
Very Slow	100.00%

Insight

Customer cancellation rates increase across the observed VTAT categories.

The **100.00% Very Slow result should be treated cautiously**, as it may reflect a small or highly structured subset of observations. The analysis therefore identifies an association rather than proving that longer arrival time causes cancellations.

6.5 Vehicle-Type Performance

Vehicle-type performance is relatively consistent.

- Completion rates range from **61.44% to 62.55%**.
- Average VTAT ranges from approximately **8.40 to 8.58**.

Insight

Vehicle type does not show a large difference in overall completion performance within this dataset. It is therefore more useful as a dashboard segmentation and filtering dimension than as a primary explanation for non-completion.

6.6 Time-Based Performance

Completion rates across broad time periods range from approximately **61.73% to 62.44%**.

Hourly completion rates range from approximately **60.88% to 63.70%**.

Insight

Overall time-based differences are relatively small. Time period remains useful for monitoring and segmentation, but it does not appear to be a major source of variation in completion performance in this dataset.

6.7 Ratings and Service Quality

Among completed rides:

Metric	Average
Driver rating	4.23
Customer rating	4.40

The correlation between driver and customer ratings is approximately **-0.001**.

Insight

There is essentially no linear relationship between the two rating measures in this dataset. Driver and customer ratings should therefore be monitored as separate indicators of service quality.

7. PostgreSQL and SQL Analysis

The cleaned dataset was loaded into:

Database: `driver_quality`

Table: `uber_data_cleaned`

SQL was used to calculate the main operational metrics, including:

- Booking outcome distribution
- Completion and non-completion rates
- Cancellation rates
- Pickup-location performance
- Driver-cancellation reasons
- VTAT performance
- Vehicle-type comparisons
- Time-based performance
- Rating statistics

Using PostgreSQL provides a consistent and repeatable analytical layer between the cleaned dataset and the Power BI dashboard.

8. Power BI Dashboard

The Power BI dashboard is designed to provide an operational view of ride-service performance.

Main dashboard components

KPI Cards

- Total Bookings
- Completion Rate
- Non-Completion Rate
- Driver Cancellation Rate
- Customer Cancellation Rate

Performance Analysis

- Booking outcome distribution

- Top pickup locations by non-completion rate
- Driver-cancellation reasons
- Customer cancellation rate by VTAT band

Interactive Filters

- Month
- Vehicle Type
- Time Period

The dashboard allows users to move from the overall booking baseline into specific pickup-location performance, cancellation patterns, and service conditions. Interactive filters can be used to examine how these metrics vary across different months, vehicle types, and time periods.

9. Recommendations

Based on the observed patterns:

1. **Investigate high non-completion locations**, particularly locations with substantially higher rates than the overall baseline.
2. **Monitor driver cancellations by location and reason** to identify operational segments with recurring cancellation activity.
3. **Track VTAT alongside customer cancellations** to identify segments where longer arrival times coincide with increased cancellation rates.
4. **Use vehicle type and time period as segmentation dimensions** when monitoring performance, while recognizing that the observed differences across these dimensions are relatively small.
5. **Monitor driver and customer ratings separately** as distinct service-quality indicators.
6. **Enrich future datasets** with driver availability, traffic, weather, pickup coordinates, driver identifiers, and incentive information to investigate operational causes more directly.

10. Limitations

- The dataset does not contain a unique driver identifier, so individual driver performance cannot be measured.
- Some missing values are structurally related to booking outcomes.
- Several distributions appear highly structured, so unusual results should be validated before being generalized.
- The analysis identifies associations rather than causal relationships.

- Traffic, weather, driver availability, and other contextual variables are not available, limiting the ability to explain why specific segments perform differently.

11. Conclusion

The analysis shows an overall **62.0% booking completion rate**, with driver cancellations representing the largest individual non-completion category. **Pickup location** shows meaningful variation in non-completion performance, while **VTAT has a strong observed association with customer cancellation rates**. In contrast, vehicle type and broad time periods show relatively limited differences.

The project demonstrates an end-to-end analytics workflow using:

Python → Data Validation & Feature Engineering → PostgreSQL → SQL Analysis → Power BI

The resulting analysis provides an operational framework for monitoring ride performance, identifying segments that warrant further investigation, and determining where additional data could support deeper analysis.